Jose Eduardo dos Santos Engineering Campus

Computer Programming I | SRS Document | Semester 1, 2026

System Requirements Specification (SRS)

MineOps

#MOTTO: Precision in the Pit. Safety in the Cloud.

Group Name: SYNTAX CREW_SRS_I3691CP
Course: Computer Programming I
Tech Stack: Expo + React Native + Firebase.
Deadline: 25 April 2026 — 23:59

Section 1 — Introduction

SYNTAX CREW'S MINEOPS APP SRS:

This pdf document presents the Software Requirements Specification for MineOps, a mobile application to be built using Expo, React Native, and Google Firebase. The application serves as a digital command centre for mining operations, synchronizing shift schedules with real-time hazard data.

Group members and their roles

Member Name	Contact Details	Student Number	Assigned Role
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1.2 Purpose of This Document

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for the MineOps mobile application. It serves as the authoritative reference document for all development, testing, and evaluation activities undertaken during the semester project. The document establishes a shared understanding between the development team, academic supervisors, and any stakeholders regarding what the system shall do and the constraints under which it shall operate.

1.3 Scope — What the App Does and Does Not Do

What MineOps Does

MineOps is a cross-platform mobile and web application (React Native + Google Firebase) targeting the following capabilities:

- Task creation, assignment, prioritisation, and tracking
- Shift scheduling and crew management
- Real-time schedule visibility and conflict detection
- Hazard and incident flagging linked to tasks
- Offline-first operation for deep pit environments with low connectivity
- Push notifications and real-time alerts for supervisors and crew

What MineOps Does Not Do:

- MineOps does not directly control or interface with physical mining machinery or IoT sensor hardware.
- MineOps does not provide payroll, HR, or procurement management functions.
- MineOps does not operate as a standalone desktop or web application; it is mobile-first (iOS and Android).

1.4 Engineering Domain & Problem Statement

MineOps addresses a critical safety and operational efficiency gap in the mining industry, situated within the domain of Industrial Safety Engineering and Mobile Information Systems.

Problem Statement (from Idea Registration Form):

Mining operations suffer from systemic communication failures during shift transitions. Paper-based hazard logs create 4+ hour dead zones where newly arriving teams are unaware of active hazards reported by the outgoing shift. Simultaneously, supervisors managing 200+ contractors lack real-time proximity data, which leads to overlapping high-risk blast zone maneuvers. MineOps eliminates these vulnerabilities by providing a single digital source of truth accessible from any Android 10+ device, synchronised via Firebase Realtime Database with sub-second propagation.

Section 2 — System Overview

2.1 System Description

MineOps is a cross-platform mobile application built with Expo and React Native, backed by Google Firebase. The system functions as a real-time operational command centre, enabling mining site personnel to manage shift schedules, submit inspection reports, raise hazard alerts, and monitor personnel proximity, all from a single mobile interface.

The application is designed for high-glare, low-connectivity environments typical of open-pit and underground mining operations. It employs an offline-first architecture, storing critical data locally on-device and synchronising with Firebase once connectivity is restored.

2.2 Target Users

- **Site Supervisors:** Primary users who manage 200+ contractors, raise hazard alerts, and review daily inspection reports. Require high-clearance data access.
- **Shift Leaders:** Responsible for shift handovers; they create end-of-shift reports and review incoming hazard logs to brief their teams.
- **Compliance Officers:** Monitor the audit trail of incidents and inspection reports to ensure MSHA/OSHA digital logging compliance.
- **Fleet Managers:** Track heavy machinery health, location, and operator fatigue indicators within the pit.
- **Mine Workers / Contractors:** Receive hazard notifications and acknowledge safety briefings through the app.

2.3 Technology Overview

- **Frontend:** Expo (managed workflow) with React Native. Provides cross-platform iOS and Android parity, native device bridges for GPS and accelerometer telemetry, and a responsive UI optimised for high-glare environments.
- **Backend:** Google Firebase — Realtime Database for sub-second data propagation, Cloud Functions for automated safety lockout logic, Firebase Authentication for IAM, and Firebase Storage for photo attachments on inspection reports.
- **Architecture Pattern:** Offline-first with local AsyncStorage caching; sync events triggered on reconnection.

2.4 Firebase Data Model

The following Firestore / Realtime Database collections and document structures will be used:

- **users/{uid}:** Stores user profile, role, site assignment, and contact details. Access controlled by Firebase Security Rules.
- **shifts/{shiftId}:** Contains shift start/end times, assigned personnel, handover notes, and status (active / closed).
- **hazards/{hazardId}:** Records hazard type, GPS coordinates, severity, reporting user, timestamp, and acknowledgement status.
- **inspections/{reportId}:** Stores daily inspection reports with zone ID, supervisor UID, photo URL (Firebase Storage), timestamp, and findings.
- **alerts/{alertId}:** Real-time broadcast documents for blast zone proximity warnings; consumed by Cloud Functions to trigger push notifications.

2.5 Assumptions and Constraints

- **Android Requirement:** The application must function on Android 10.0 (API Level 29) and above. iOS support is secondary.
- **Internet Dependency:** Firebase Realtime sync requires an active internet connection. All critical data is cached locally to ensure continuity during outages.
- **GPS Dependency:** Location-sensitive features (proximity alerts, blast zone warnings) require the device's GPS to be enabled.
- **Firebase Project:** All backend services are provisioned within a single Firebase project. Cloud Functions require a Blaze (pay-as-you-go) plan.
- **Device Capability:** Minimum 2 GB RAM and a rear-facing camera are required for inspection photo capture.

Section 3 — Functional Requirements

Each requirement has a unique ID and references the Firebase service it involves. Each requirement must be specific and testable. Minimum 10 functional requirements are required.

ID	Requirement Statement	Priority	Firebase Service
FR-001	The system shall allow a new user to register with an email address and password using Firebase Authentication.	**High**	Firebase Auth
FR-002	The system shall allow a registered user to log in and log out securely using Firebase Authentication.	**High**	Firebase Auth
FR-003	The system shall allow a mine supervisor to create a daily inspection report saved to Firestore.	**High**	Firestore
FR-004	The system shall display all inspection reports submitted by the logged-in user in date order.	**High**	Firestore
FR-005	The system shall allow a user to attach a photo to a report, stored in Firebase Storage.	**Medium**	Firebase Storage
FR-006	The system shall broadcast a real-time hazard alert to all users assigned to an active shift when a new hazard is created.	**High**	Realtime Database
FR-007	The system shall allow a shift leader to create and close a shift handover record, capturing outgoing and incoming team members.	**High**	Firestore
FR-008	The system shall enforce role-based access control, preventing standard workers from accessing supervisor-level reports.	**High**	Firebase Auth / Rules
FR-009	The system shall trigger a Cloud Function to send a push notification when a hazard is raised within a geofenced blast zone.	**High**	Cloud Functions
FR-010	The system shall display a live HUD showing the GPS coordinates and status of all active personnel on an active shift.	**Medium**	Realtime Database
FR-011	The system shall cache all unsynced inspection reports locally and automatically upload them when connectivity is restored.	**High**	AsyncStorage / Firestore
FR-012	The system shall allow a compliance officer to export a PDF audit trail of all incidents logged within a specified date range.	**Medium**	Cloud Functions / Storage

Section 4 — Non-Functional Requirements

- **Performance:** The app shall load the main dashboard within 3 seconds on a standard Wi-Fi connection.
- **Security:** All Firestore rules shall require request.auth != null for any read or write operation. AES-256 encryption is applied at rest.
- **Usability:** A new user shall be able to complete registration without external guidance; no more than 4 taps to raise a hazard alert.
- **Compatibility:** The app shall function on Android 10.0 (API Level 29) and above on devices with a minimum of 2 GB RAM.
- **Reliability:** Locally entered data shall not be lost if the device briefly loses connectivity; offline-first caching must preserve data integrity.
- **Availability:** Firebase backend services shall maintain 99.9% uptime as per Google Firebase SLA.
- **Scalability:** The architecture shall support at minimum 500 concurrent users across multiple mine sites without degradation.

Section 5 — Use Case Diagrams

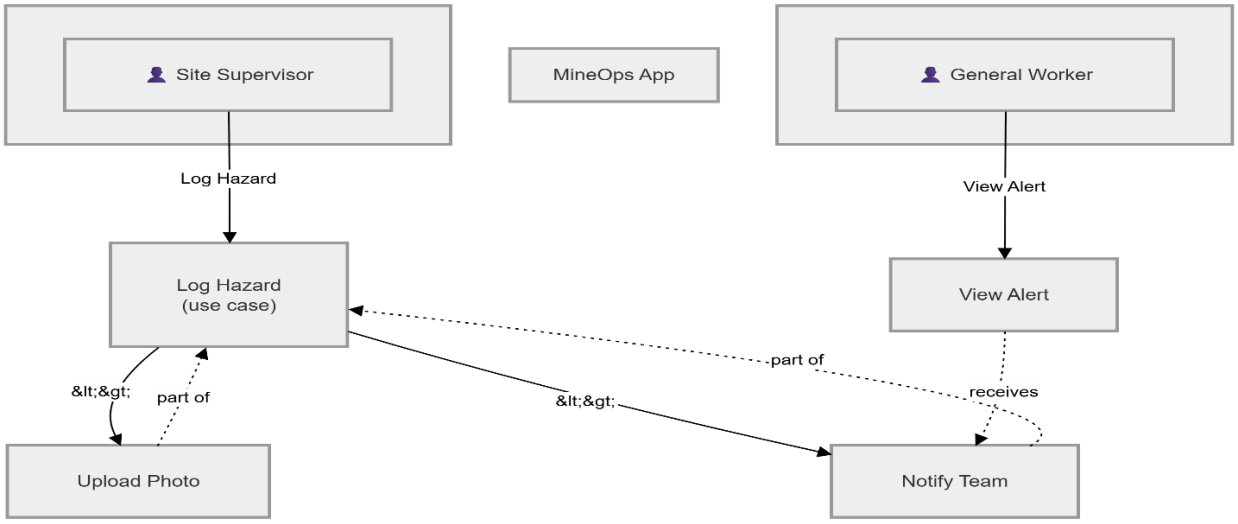
This section visualizes the interaction between the primary actors (Site Supervisor, Fleet Manager, Compliance Officer) and the MineOps system. The diagrams illustrate the core functionalities defined in the Functional Requirements (Section 3).

5.1 Actors Description

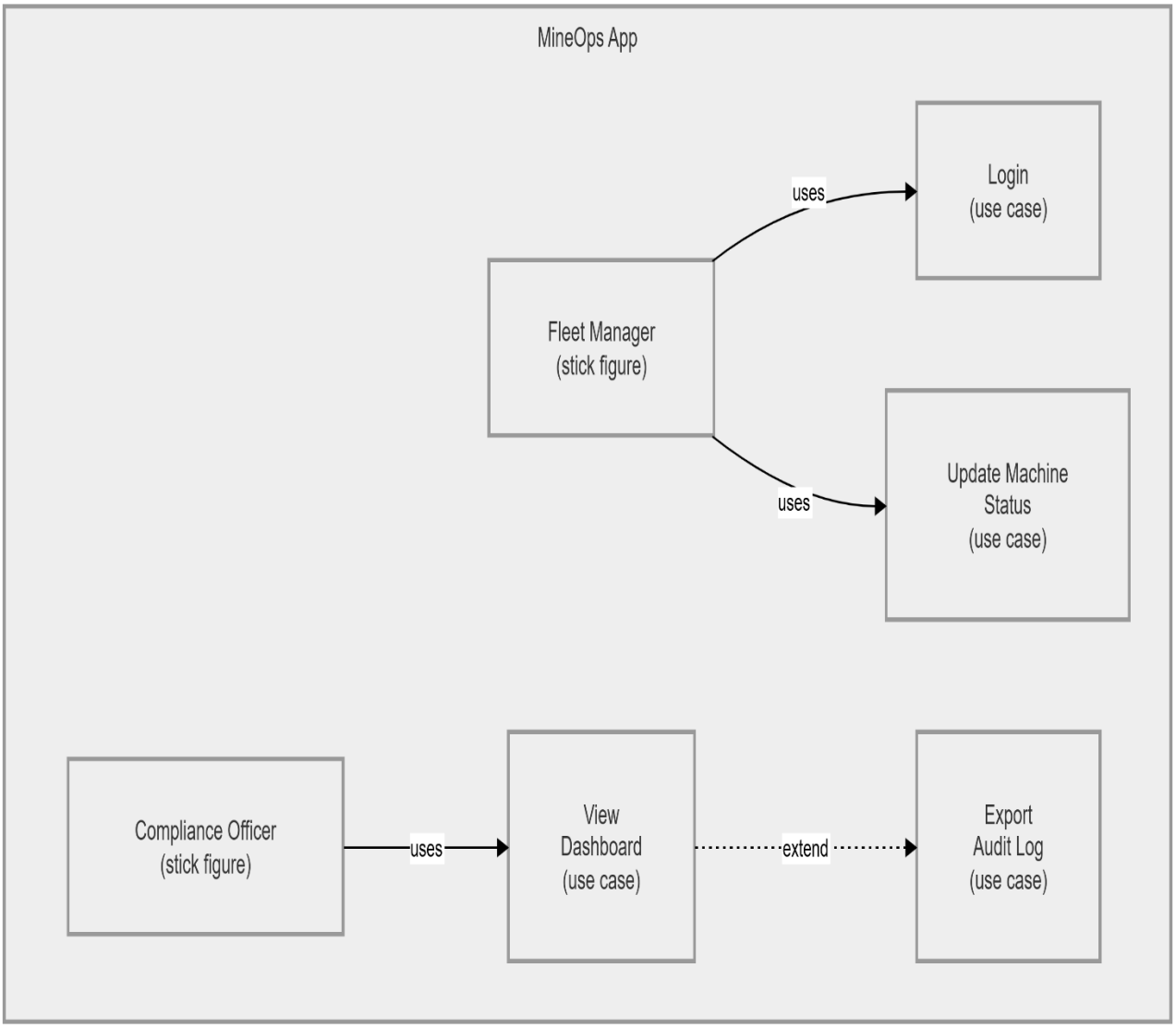
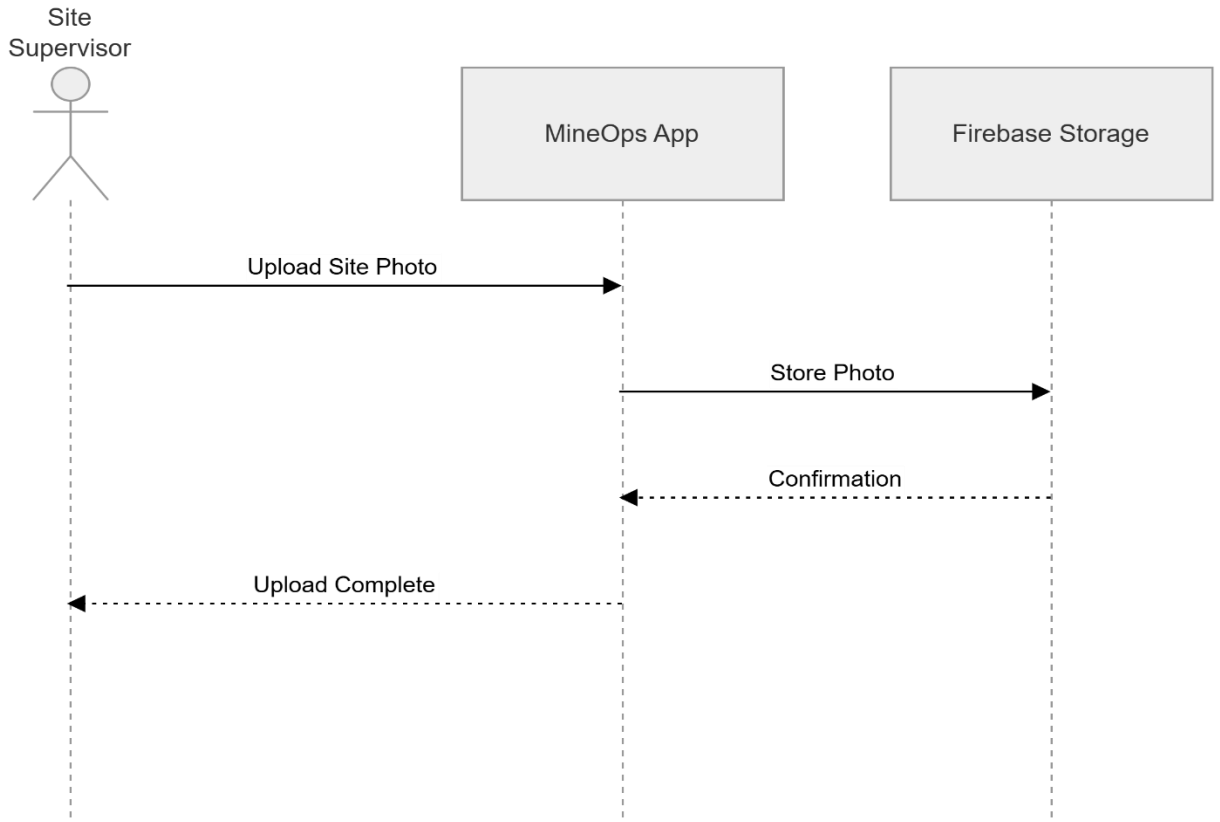
Actor	Description
Site Supervisor	The primary user responsible for logging hazards, managing shift handovers, and verifying safety protocols in real-time.
Fleet Manager	Responsible for maintaining the digital status of heavy machinery and updating maintenance logs.
Compliance Officer	A read-only auditor who reviews safety logs, exports reports for external audits, and monitors "Blast Zone" alerts.
General Worker	Uses the system primarily to view safety notifications, personal shift schedules, and "Blast Zone" warnings.

5.2 Case Descriptions

Case Diagram 1



Case Diagram 2



Case Diagram 3